

1 Intrinsic convergence on the completed carrier

Source: PrimeTensor/Analysis/Limit.lean

What this file establishes

MulReal has so far been an algebraic object: a commutative carrier in the sense of PrimeTensor/Structure.lean, receiving a structure-preserving map from MulRat. Nothing has related its elements to the scale they were built from. This file supplies the missing analytic vocabulary — a nearness relation on completed points and a notion of convergence for sequences of them — and proves the one theorem that makes the construction deserve the name completion:

$$\text{for every Cauchy stream } a, \quad \text{ofRat}(a_n) \longrightarrow [a].$$

That is, the embedded rational terms of a Cauchy stream converge to the point that stream represents. Read the other way, it says that `ofRat(MulRat)` is dense in `MulReal`: every completed point is a limit of embedded rationals, with an explicit sequence exhibiting it.

The design question the file has to answer is how to transport a relation defined on representatives across a quotient whose relation is not itself the one being transported. Its answer — quantify existentially over representatives — is what Design note 1.2 is about, and it is what makes the central theorem cost nothing.

1.1 Nearness of completed points

Definition 1.1 (`MulReal.ScaleNear`). Completed points x and y are *near at level ℓ* when there exist Cauchy streams a and b with $[a] = x$ and $[b] = y$, and an anchor beyond which their terms are `ScaleWithin( $\ell, -, -$ )`.

```
def ScaleNear (level : Depth) (x y : MulReal) : Prop :=
  ∃ a b : MulCauchyStream,
    ofStream a = x ∧
    ofStream b = y ∧
    ∃ anchor : Depth,
      ∀ n : Depth,
        Depth.AtOrAfter anchor n →
          MulRat.ScaleWithin level (a.term n) (b.term n)
```

Design note 1.2 (Existential, not universal, over representatives). A relation on representatives descends to a quotient when it is invariant under the quotient’s own relation. `ScaleWithin` at a fixed level is not: two streams asymptotic in the sense of PrimeTensor/Completion.lean agree only up to the composition law of PrimeTensor/Scale/Laws.lean, and that law consumes a level each time it is used. Replacing representatives therefore cannot be free at a fixed level, and a relation defined by “*for all* representatives” and one defined by “*there exist* representatives” are genuinely different propositions here, the second being the weaker.

The file takes the weaker one deliberately — the docstring calls the choice part of the quotient semantics. The consequence, visible immediately below, is that establishing `ScaleNear` means exhibiting one convenient pair of representatives rather than controlling every pair, and the central theorem gets to choose the pair that makes it trivial. The price is paid in what can be proved *about* the relation, and Remark 1.4 is the bill.

Note also what is not introduced. There is no metric, no distance function, and no additive ε : nearness is a family of relations indexed by depth, exactly as at the MulRat layer, and the index is the same Depth that indexes the sequences.

Lemma 1.3 (MulReal.scaleNear_refl and scaleNear_symm). *ScaleNear(ℓ, x, x) for every level and every x ; and ScaleNear(ℓ, x, y) implies ScaleNear(ℓ, y, x).*

Proof. For reflexivity, induct to choose a representative a of x and offer the pair (a, a) , with anchor one so the tail is everything; each term comparison is scaleWithin_refl. For symmetry, take the witnessing pair supplied by the hypothesis, exchange its two components together with the two identifications, keep the anchor and the level, and apply scaleWithin_symm termwise. $\square$

Remark 1.4 (Transitivity is not proved, at any level). ScaleWithin was not transitive at a fixed level either, but it had the level-shifted substitute of PrimeTensor/Scale/Laws.lean. No analogue is proved here, and the obstruction is exactly the existential of Design note 1.2: a witness for ScaleNear(ℓ, x, y) names some representative of y , a witness for ScaleNear(ℓ, y, z) names *some other* one, and the two need not be the same stream. Chaining them requires first crossing from one representative of y to the other, which is possible — they are asymptotic, so they are eventually close at every level — but costs a further application of the composition law, and then a comparison between two different levels, which is the monotonicity statement that PrimeTensor/Scale/Laws.lean left unproved. So a level-shifted transitivity is plausible and is not available off the shelf. Nothing below needs it.

1.2 Convergence

Definition 1.5 (MulReal.Seq and MulReal.ConvergesTo). A *sequence* is a function $s : \text{Depth} \rightarrow \text{MulReal}$, and s *converges to* x when

$$\forall \ell \exists c \forall n : \text{AtOrAfter}(c, n) \longrightarrow \text{ScaleNear}(\ell, s_n, x).$$

```
def ConvergesTo (s : Seq) (x : MulReal) : Prop :=
  ∀ level : Depth,
    ∃ anchor : Depth,
      ∀ n : Depth,
        Depth.AtOrAfter anchor n →
          ScaleNear level (s n) x
```

The shape is the one used twice already — for IsMulCauchy and for MulAsymptotic in PrimeTensor/Completion.lean — with ScaleNear in place of ScaleWithin: for every level a tail on which the requirement holds, with the level and the tail anchor both depths and nothing else quantified.

Lemma 1.6 (MulReal.converges_constant). *The constant sequence at x converges to x .*

Proof. Anchor one, and each requirement is Lemma 1.3. $\square$

Lemma 1.7 (MulReal.ofStream_eq_of_asymptotic). *Asymptotic streams represent the same completed point.*

Proof. This is Quotient.sound for the setoid of PrimeTensor/Completion.lean, named so that later proofs can rewrite along it. $\square$

1.3 The convergence theorem

Theorem 1.8 (`MulReal.terms_converge`). *For every Cauchy stream a , the sequence $n \mapsto \text{ofRat}(a_n)$ converges to $[a]$.*

```
theorem terms_converge (a : MulCauchyStream) :
  ConvergesTo (fun n => ofRat (a.term n)) (ofStream a) := by
  intro level
  obtain (anchor, hcauchy) := a.cauchy level
  refine (anchor, ?_)
  intro n hn
  refine (MulCauchyStream.constant (a.term n), a, rfl, rfl, anchor, ?_)
  intro k hk
  exact hcauchy n k hn hk
```

Proof. Fix a level ℓ and let c be the anchor supplied by the Cauchy property of a at that same level — not at $\text{succ } \ell$. Take c as the anchor for the convergence claim as well.

Let n be at or after c . To witness $\text{ScaleNear}(\ell, \text{ofRat}(a_n), [a])$, offer the pair consisting of the constant stream at a_n and the stream a itself. The two identifications are `rfl`: $\text{ofRat}(q)$ was defined in `PrimeTensor/Completion.lean` as the class of the constant stream at q , and the second component is a unchanged. Take c once more as the anchor of the witness. For k at or after c , the required comparison is

$$\text{ScaleWithin}(\ell, (\text{constant}(a_n))_k, a_k) = \text{ScaleWithin}(\ell, a_n, a_k),$$

and both n and k lie in the tail at c , so this is precisely the Cauchy property of a at ℓ applied to the pair (n, k) . $\square$

Remark 1.9 (The proof costs nothing, and why). No level is spent: the Cauchy property is invoked at ℓ and the conclusion is at ℓ . No join of anchors appears; one anchor serves for the convergence claim and for every witness. The whole theorem is the Cauchy property of a , re-read.

That is the payoff of Design note 1.2. The witnessing pair may be chosen, and the choice made here — compare the constant stream at a_n against a itself, rather than against some other representative — turns the statement “the n -th embedded term is near the limit” into the statement “ a_n is close to a_k for large k ”, which is what being Cauchy says. Under a universally quantified ScaleNear the same argument would have to survive an arbitrary change of representative on both sides, and would spend levels doing it.

Corollary 1.10 (`MulReal.terms_converge_of_asymptotic`). *If a and b are asymptotic then $n \mapsto \text{ofRat}(a_n)$ converges to $[b]$.*

Proof. $[a] = [b]$ by Lemma 1.7; rewrite the target and apply Theorem 1.8. $\square$

Remark 1.11 (Density, and the half of completeness this is). Theorem 1.8 says every completed point is a limit of embedded rationals, and it does more than assert existence: the approximating sequence is exhibited, being the terms of any representative. That is the density half of the statement that `MulReal` completes `MulRat`.

The other half — that `MulReal` is itself complete, every Cauchy sequence of completed points converging in `MulReal` — is not proved, and cannot yet be stated: `IsMulCauchy` is a predicate on `MulRat`-valued streams, and no Cauchy condition for a `MulReal`-valued `Seq` is defined anywhere in the file. The vocabulary for the second half is missing, not merely its proof.

Remark 1.12 (Limits are not yet unique). `ConvergesTo` is a relation, and nothing here makes it a function. If s converges to x and to y , no lemma concludes $x = y$; the usual argument would combine the two through a transitivity that Remark 1.4 says is unavailable, and would then need the separation statement that two points near at every level are equal. That statement is the same missing ingredient as the injectivity of `ofRat`, noted at the end of `PrimeTensor/Structure.lean`: both are the assertion that the ladder of levels actually separates points. Until it is proved one may not speak of *the* limit of a sequence, and this file never does.

Summary of the correspondence

Lean declaration	Kind	Here
<code>MulReal.ScaleNear</code>	definition	Def. 1.1
<code>MulReal.scaleNear_refl</code>	theorem	Lem. 1.3
<code>MulReal.scaleNear_symm</code>	theorem	Lem. 1.3
<code>MulReal.Seq</code>	abbrev	Def. 1.5
<code>MulReal.ConvergesTo</code>	definition	Def. 1.5
<code>MulReal.converges_constant</code>	theorem	Lem. 1.6
<code>MulReal.ofStream_eq_of_asymptotic</code>	theorem	Lem. 1.7
<code>MulReal.terms_converge</code>	theorem	Thm. 1.8
<code>MulReal.terms_converge_of_asymptotic</code>	theorem	Cor. 1.10

Every declaration of `PrimeTensor/Analysis/Limit.lean` appears above. The file contains no `sorry`, no axiom, and no unproved named proposition. Remarks 1.4, 1.9, 1.11 and 1.12 are stated for the reader and are not declarations of the file.